2.2 Verifying Trigonometric Identities

Name _____ Period ____ Date _____

Pre-Calculus 2.2 (30-minute plan) · Student sheet

I can…

- prove a trig identity by working one side until it matches the other.

Vocabulary · fill in the missing word

Identity — an _____ that is true for every value in its domain

Conditional equation — _____ only for some values (those are the solutions)

Verify — show the _____ is true for EVERY x, working one side at a time

Conjugate — 1 + sin x and 1 − sin x; _____ them gives $1 - \sin^2 x = \cos^2 x$

Common denominator — used to add or subtract trig _____

Part 1 · Notes and examples (fill in each note, then work the example)

Guideline 1: 1. Work with _____ side at a time. Start with the _____ side.

Guideline 2: 2. Look for algebra: _____, add fractions, square a binomial, or multiply by a _____.

Guideline 3: 3. Look for places to use the _____ identities.

Careful: When verifying, never move terms across the _____ sign or do the same thing to both sides.

Example 1

Verify: $\dfrac{\sec^2\theta - 1}{\sec^2\theta} = \sin^2\theta$

Guideline 4: 4. If nothing else works, convert everything to _____ and _____.

Example 2

Verify: $\dfrac{1}{1-\sin\alpha} + \dfrac{1}{1+\sin\alpha} = 2\sec^2\alpha$

Part 2 · Partner practice

Partner practice

Verify: $\tan x + \cot x = \sec x \csc x$

Part 3 · On your own: choose 2 of 3

Problem 1

Verify: $\cos x(\sec x - \cos x) = \sin^2 x$

Problem 2

Verify: $\csc x - \sin x = \cos x \cot x$

Problem 3

Verify: $\dfrac{\sin x}{1 - \cos x} = \dfrac{1 + \cos x}{\sin x}$